AI ENGINEER ROADMAP | ├── 1. Programming Fundamentals | ├── Python | | ├── Variables & Data Types | | ├── Functions & Modules | | ├── OOP & Exception Handling | | ├── List & Dictionary Comprehensions | | ├── Iterators & Generators | | ├── Decorators & Context Managers | | ├── Asyncio & Concurrency | | ├── Type Hints & Pydantic | | └── Virtual Environments & Packaging | ├── Data Structures & Algorithms | ├── Git & GitHub | ├── Linux & Command Line | ├── HTTP & Networking Fundamentals | ├── REST APIs | └── JSON, YAML & Serialization | ├── 2. Mathematics & ML Foundations | ├── Linear Algebra | | ├── Vectors & Matrices | | ├── Dot Products | | ├── Matrix Multiplication | | └── Eigenvalues & Eigenvectors | ├── Calculus | | ├── Derivatives | | ├── Partial Derivatives | | ├── Gradients | | └── Chain Rule | ├── Probability & Statistics | | ├── Probability Distributions | | ├── Bayes' Theorem | | ├── Mean & Variance | | └── Statistical Inference | ├── Machine Learning Fundamentals | | ├── Supervised & Unsupervised Learning | | ├── Training & Testing | | ├── Overfitting & Underfitting | | ├── Loss Functions | | └── Model Evaluation | └── Deep Learning Fundamentals | ├── Neural Networks | ├── Backpropagation | ├── Gradient Descent | └── Activation Functions | ├── 3. Data Engineering & Processing | ├── NumPy | ├── Pandas | ├── Data Cleaning | ├── Data Transformation | ├── Exploratory Data Analysis | ├── SQL & PostgreSQL | ├── JSON & JSONL | ├── CSV, XML & Parquet | ├── Data Pipelines | ├── Web Scraping | ├── Data Validation | └── Dataset Preparation | ├── 4. Deep Learning & Neural Networks | ├── Perceptrons | ├── Feedforward Neural Networks | ├── Forward & Backpropagation | ├── Loss Functions | ├── Optimizers | ├── CNNs & RNNs | ├── LSTMs & GRUs | ├── Attention Mechanisms | ├── Transformer Architecture | ├── Embeddings | └── Transfer Learning | ├── 5. Deep Learning Frameworks | ├── PyTorch | | ├── Tensors | | ├── Autograd | | ├── nn.Module | | ├── DataLoaders | | ├── Custom Training Loops | | ├── GPU Training | | └── Model Checkpointing | ├── TensorFlow & Keras Basics | ├── Hugging Face Ecosystem | | ├── Transformers | | ├── Datasets | | ├── Tokenizers | | └── Accelerate | └── CUDA Fundamentals | ├── 6. LLM Fundamentals | ├── What Are Large Language Models? | ├── Transformer Architecture | ├── Encoder & Decoder Models | ├── GPT Architecture | ├── BERT Architecture | ├── Tokenization | ├── Vocabulary & Token IDs | ├── Context Windows | ├── Positional Encoding | ├── Self-Attention & Multi-Head Attention | ├── Pretraining & Post-Training | ├── Instruction Tuning | ├── RLHF & DPO | ├── Model Parameters | └── Open-Source vs Proprietary Models | ├── 7. LLM APIs & Model Integration | ├── OpenAI API | ├── Anthropic API | ├── Google Gemini API | ├── Open-Weight Models | ├──

API Authentication | ├── Streaming Responses | ├── Structured Outputs | ├── Function & Tool Calling | ├── Multimodal APIs | ├── Error Handling & Retries | ├── Rate Limits | ├── Token Usage & Cost Tracking | └── Model Routing & Fallbacks | ├── 8. Prompt & Context Engineering | ├── System & User Instructions | ├── Few-Shot Prompting | ├── Chain-of-Thought Concepts | ├── Structured Prompting | ├── Prompt Templates | ├── Context Window Management | ├── Context Compression | ├── Context Selection & Retrieval | ├── Prompt Versioning | ├── Prompt Injection Defense | ├── Context Engineering | └── Prompt Evaluation | ├── 9. Embeddings & Vector Databases | ├── What Are Embeddings? | ├── Dense & Sparse Embeddings | ├── Embedding Models | ├── Similarity Metrics | ├── Cosine Similarity | ├── Semantic Search | ├── Vector Indexing | ├── Approximate Nearest Neighbors | ├── Vector Databases | | ├── Pinecone | | ├── Qdrant | | ├── Weaviate | | ├── Milvus | | └── Chroma | ├── Metadata Filtering | └── Hybrid Search | ├── 10. Retrieval-Augmented Generation (RAG) | ├── RAG Architecture | ├── Document Ingestion | ├── Document Parsing | ├── Text Chunking Strategies | ├── Overlapping Chunks | ├── Embedding Generation | ├── Vector Storage | ├── Query Transformation | ├── Retrieval Strategies | ├── Reranking | ├── Hybrid Retrieval | ├── Contextual Retrieval | ├── Parent-Child Retrieval | ├── Multi-Query Retrieval | ├── Graph RAG | ├── Agentic RAG | ├── RAG Evaluation | └── Advanced RAG Optimization | ├── 11. AI Agents & Agentic Systems | ├── What Are AI Agents? | ├── Agent Architecture | ├── Reasoning & Planning | ├── Tool Calling | ├── Function Execution | ├── Short-Term & Long-Term Memory | ├── Agent State Management | ├── Workflow Orchestration | ├── Single-Agent Systems | ├── Multi-Agent Systems | ├── Human-in-the-Loop | ├── Reflection & Self-Correction | ├── Planning & Execution Loops | ├── Agent Reliability | ├── Agent Evaluation | └── Agent Security | ├── 12. AI Frameworks & Orchestration | ├── LangChain | ├── LangGraph | ├── LlamaIndex | ├── Haystack | ├── Semantic Kernel | ├── Pydantic AI | ├── DSPy | ├── CrewAI | ├── AutoGen | ├── Workflow Orchestration | └── Framework Selection & Tradeoffs | ├── 13. Model Fine-Tuning & Adaptation | ├── Pretrained Models | ├── Supervised Fine-Tuning (SFT) | ├── Parameter-Efficient Fine-Tuning | ├── LoRA | ├── QLoRA | ├── PEFT | ├── Dataset Preparation | ├── Synthetic Data Generation | ├── Instruction Tuning | ├── DPO | ├── Fine-Tuning Evaluation | └── When to Fine-Tune vs RAG | ├── 14. Multimodal AI Engineering | ├── Vision-Language Models | ├── Image Understanding | ├── OCR & Document Intelligence | ├── Speech-to-Text | ├── Text-to-Speech | ├── Audio Understanding | ├── Video Understanding | ├── Multimodal Embeddings | ├── Image Generation Models | ├── Diffusion Models |

└── Multimodal RAG | ├── 15. AI Evaluation & Testing | ├── Why AI Evaluation Matters | ├── Unit & Integration Testing | ├── LLM-as-a-Judge | ├── Golden Datasets | ├── Evaluation Datasets | ├── Model Accuracy | ├── Faithfulness & Groundedness | ├── Relevance & Context Precision | ├── Hallucination Detection | ├── RAG Evaluation | ├── Agent Evaluation | ├── Human Evaluation | ├── Regression Testing | ├── Online & Offline Evaluation | └── Evals & Observability | ├── 16. AI Infrastructure & Inference Engineering | ├── LLM Inference Fundamentals | ├── Prefill vs Decode | ├── KV Cache Management | ├── Continuous Batching | ├── PagedAttention | ├── Speculative Decoding | ├── Quantization | ├── Model Pruning | ├── Knowledge Distillation | ├── Model Serving | ├── vLLM | ├── SGLang | ├── Ollama | ├── TensorRT-LLM | ├── GPU Memory Management | ├── Latency & Throughput Optimization | └── Distributed Inference Basics | ├── 17. Backend Engineering for AI | ├── FastAPI | ├── Flask | ├── REST APIs | ├── WebSockets & Streaming | ├── Async Processing | ├── Redis & Caching | ├── PostgreSQL | ├── Background Jobs | ├── Message Queues | ├── Authentication & Authorization | ├── Rate Limiting | ├── API Security | ├── Microservices Basics | └── AI Application Architecture | ├── 18. MLOps & LLMOps | ├── ML & LLM Lifecycle Management | ├── Experiment Tracking | ├── MLflow | ├── Weights & Biases | ├── Langfuse | ├── LangSmith | ├── Prompt Versioning | ├── Model Versioning | ├── Data Versioning (DVC) | ├── Tracing & Observability | ├── Cost Monitoring | ├── Model Monitoring | ├── CI/CD for AI Applications | └── Automated Evaluation Pipelines | ├── 19. Cloud & DevOps | ├── AWS | | ├── S3 | | ├── EC2 | | ├── Lambda | | ├── ECS & EKS | | └── SageMaker Basics | ├── Google Cloud & Vertex AI | ├── Microsoft Azure AI | ├── Docker | ├── Kubernetes | ├── GitHub Actions | ├── CI/CD Pipelines | ├── Terraform | ├── GPU Cloud Infrastructure | └── Deployment & Scaling | ├── 20. AI Security & Responsible AI | ├── Prompt Injection | ├── Jailbreak Attacks | ├── Data Leakage | ├── Sensitive Data Protection | ├── PII Detection & Redaction | ├── Tool Execution Security | ├── Sandboxed Execution | ├── Model Supply Chain Security | ├── Adversarial Attacks | ├── Bias & Fairness | ├── Explainability | ├── AI Governance | └── Guardrails & Content Safety | ├── 21. AI System Design & Architecture | ├── Designing Scalable AI Applications | ├── RAG System Design | ├── Agent System Design | ├── Model Routing Architecture | ├── Multi-Tenant AI Systems | ├── Caching Strategies | ├── Database & Vector Storage Design | ├── Distributed Task Processing | ├── Fault Tolerance & Retries | ├── Cost Optimization | ├── Latency Optimization | ├── Horizontal Scaling | ├── Reliability & Availability | └── AI Tradeoff Analysis | ├── 22. Build Real-World Projects | ├── AI-Powered

Research Assistant | ├── Document Intelligence Platform | ├── Production-Ready RAG System | ├── Multi-Agent Research System | ├── AI Customer Support Agent | ├── AI Coding Assistant | ├── Multimodal AI Assistant | ├── AI Meeting Intelligence System | ├── Enterprise Knowledge Assistant | ├── AI Workflow Automation Platform | ├── Custom LLM Fine-Tuning Pipeline | ├── AI Model Evaluation Platform | └── End-to-End Production AI SaaS | └── 23. Career Preparation ├── Build a Strong GitHub Portfolio ├── Create an AI Engineering Resume ├── Master Python & SQL Interviews ├── Learn ML & LLM Fundamentals ├── Practice AI System Design ├── Understand RAG & Agent Architectures ├── Learn Inference Optimization ├── Study Production AI Case Studies ├── Contribute to Open Source ├── Write Technical Articles ├── Build in Public └── Apply for AI Engineer Roles

FINAL GOAL: Become a Production-Ready AI Engineer

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